

AWS DevOps Engineer Roadmap (Beginner to Expert)

Phase 1: Foundation (Weeks 1-4)

Goal: Understand the basics of AWS, Linux, and DevOps principles.

Key Topics:

- What is DevOps?
- Basics of Cloud Computing
- AWS Global Infrastructure (Regions, AZs)
- IAM (Users, Roles, Policies)
- EC2, S3, VPC basics
- Linux Basics & Shell Scripting
- Version Control: Git & GitHub

Hands-on:

- Create AWS Free Tier account
- Launch an EC2 instance
- Use S3 to store files
- Practice Git: clone, commit, push, pull

Resources:

- AWS Cloud Practitioner Essentials (Free)
- Codecademy/Linux Journey for Linux

[] Completed

Phase 2: Core AWS & DevOps Tools (Weeks 5-8)

Goal: Get comfortable with core AWS services and DevOps tools.

Key Topics:

- AWS CLI & SDKs
- EC2 AMIs, Security Groups, Key Pairs
- Auto Scaling & Load Balancers
- CloudWatch & CloudTrail
- CI/CD concepts
- Jenkins: Setup, Pipelines
- Docker: Containers & Images

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- ECR & ECS basics

Hands-on:

- Install & configure Jenkins on EC2
- Write a Jenkins pipeline for a sample app
- Dockerize a simple Node.js/Spring Boot app
- Push Docker images to ECR
- Deploy to ECS

Certification:

- AWS Certified Solutions Architect - Associate (Start prep)

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Phase 3: Infrastructure as Code (Weeks 9-12)

Goal: Automate infrastructure and deployments.

Key Topics:

- Terraform (IaC): Providers, Resources, Modules
- AWS VPC, Subnets, Route Tables
- EC2, RDS, S3, IAM using Terraform
- Ansible basics (optional)
- AWS CodePipeline, CodeBuild, CodeDeploy

Hands-on:

- Write Terraform code to create an EC2 instance
- Deploy an app using CodePipeline & CodeDeploy
- Automate EC2 & RDS provisioning with Terraform

Certification:

- AWS Certified DevOps Engineer - Associate (Start prep)

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Phase 4: Containers & Orchestration (Weeks 13-16)

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Goal: Learn container orchestration and scaling apps.

Key Topics:

- Kubernetes (k8s) Concepts
- EKS (Elastic Kubernetes Service)
- Helm for packaging
- CI/CD with Jenkins + EKS
- GitOps (ArgoCD or Flux)

Hands-on:

- Launch EKS cluster using Terraform or eksctl
- Deploy app to EKS
- Use Helm to manage k8s apps
- Set up GitOps with ArgoCD

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Phase 5: Advanced DevOps & Monitoring (Weeks 17-20)

Goal: Deep dive into advanced topics and monitoring.

Key Topics:

- Monitoring & Logging (CloudWatch, Prometheus, Grafana)
- Secrets Management (AWS Secrets Manager, Parameter Store)
- Cost Optimization
- Blue/Green & Canary Deployments
- Security Best Practices (IAM, VPC, GuardDuty)
- DevSecOps basics

Hands-on:

- Monitor apps with Prometheus + Grafana
- Store app secrets securely
- Set up Blue/Green deployment with CodeDeploy

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